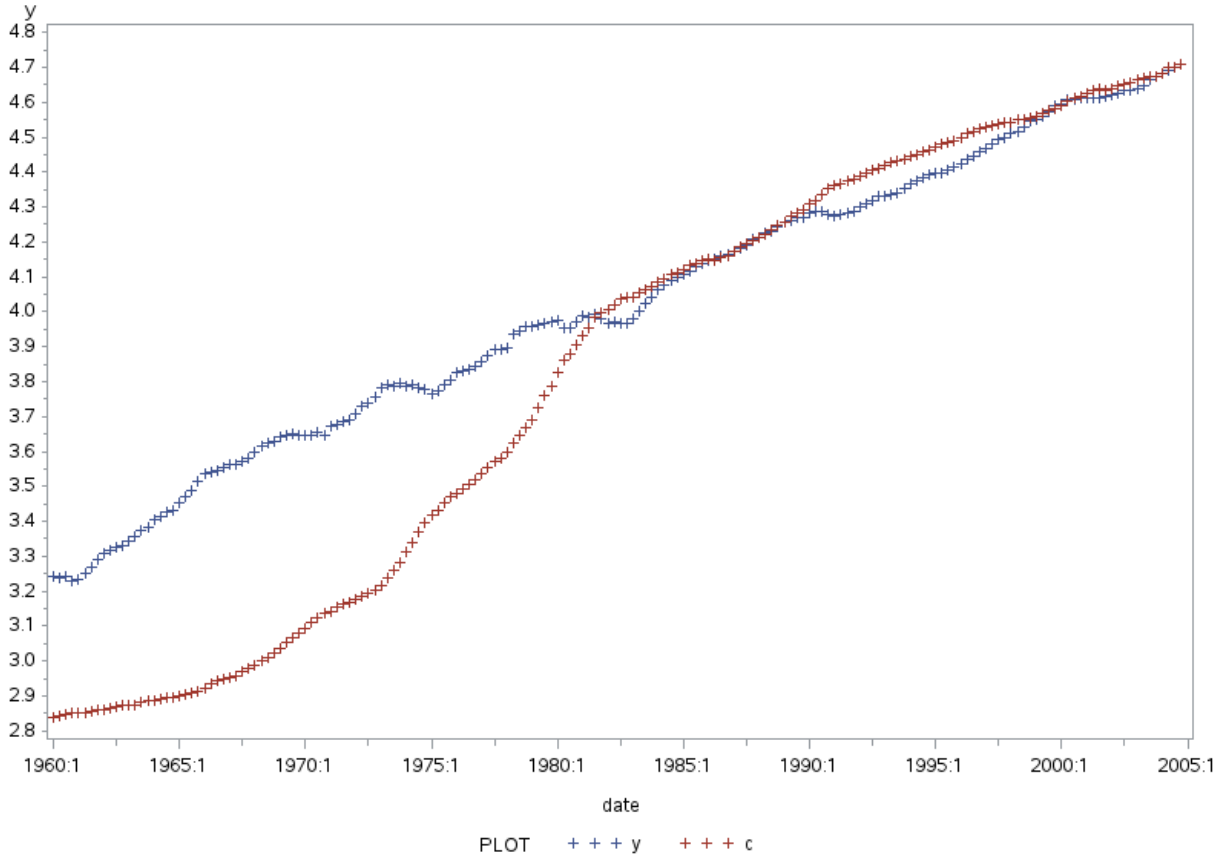
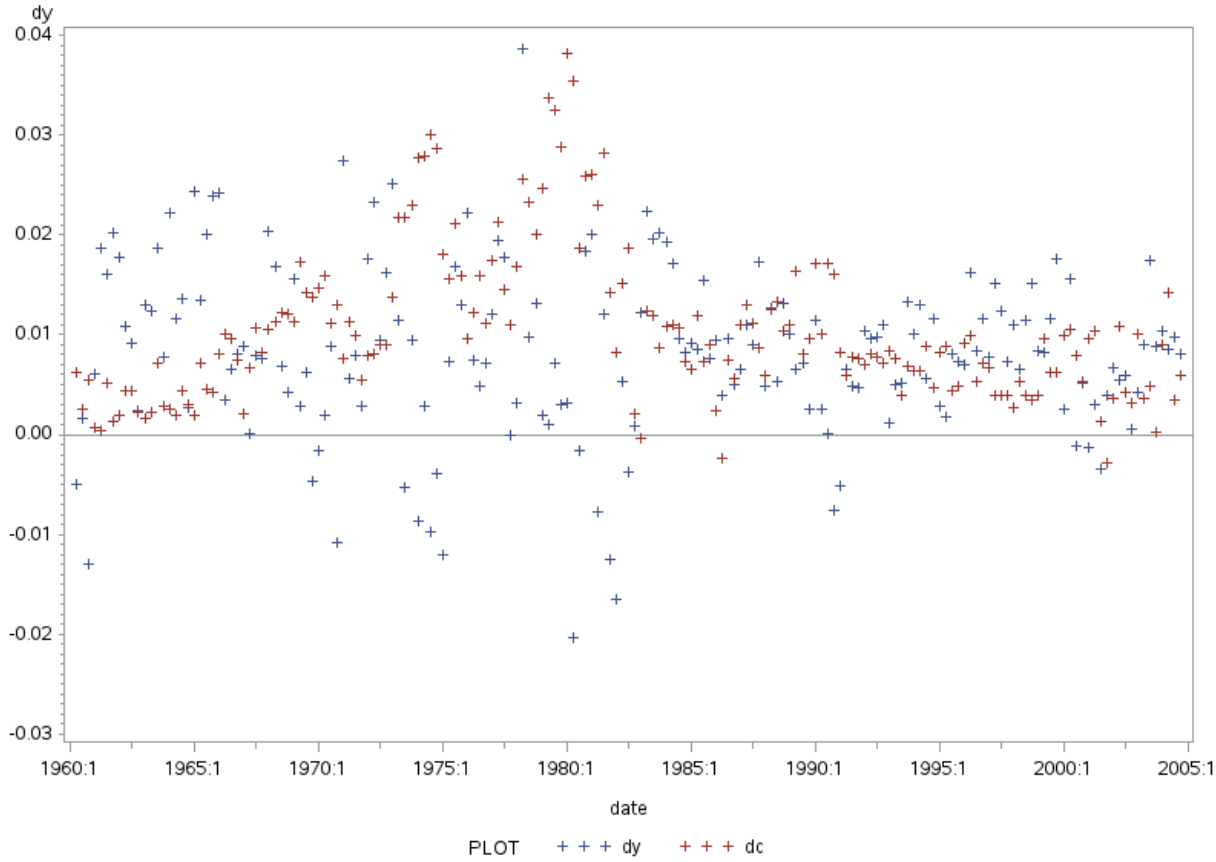
Real personal disposable income and consumption expenditure



Differences in real income and consumption expenditure



ADF test for consumption with 3 lags

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	AR_V1	AR_V2	AR_V3	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		.000016128	0.003085835	-1	-.000523892	0.68485	-0.23052	0.44586	176	-1.10041	1	1	0.71550
2	COV	0 Converged	AR_V	Intercept	.000016128	0.000003804	.	-.000000892	-0.00002	-0.00000	-0.00001	176	-1.10041	1	1	0.71550
3	COV	0 Converged	AR_V	DLAG_V	.000016128	-.000000892	.	0.000000227	0.00000	-0.00000	0.00000	176	-1.10041	1	1	0.71550
4	COV	0 Converged	AR_V	AR_V1	.000016128	-.000017644	.	0.000002371	0.00470	-0.00343	-0.00046	176	-1.10041	1	1	0.71550
5	COV	0 Converged	AR_V	AR_V2	.000016128	-.000001669	.	-.000000362	-0.00343	0.00725	-0.00354	176	-1.10041	1	1	0.71550
6	COV	0 Converged	AR_V	AR_V3	.000016128	-.000008441	.	0.000000128	-0.00046	-0.00354	0.00478	176	-1.10041	1	1	0.71550

ADF test for disposable income with 0 lags

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		.000073525	0.016485	-1	-.002069012	179	-1.32965	1	1	0.61605
2	COV	0 Converged	AR_V	Intercept	.000073525	0.000039	.	-.000009712	179	-1.32965	1	1	0.61605
3	COV	0 Converged	AR_V	DLAG_V	.000073525	-0.000010	.	0.000002421	179	-1.32965	1	1	0.61605

ADF test for differenced consumption with 3 lags

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	AR_V1	AR_V2	AR_V3	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		.000015926	0.001196958	-1	-0.10801	-0.15394	-0.41960	0.10429	175	-2.49861	1	1	0.11764
2	COV	0 Converged	AR_V	Intercept	.000015926	0.000000299	.	-0.00002	0.00001	0.00001	0.00001	175	-2.49861	1	1	0.11764
3	COV	0 Converged	AR_V	DLAG_V	.000015926	-.000019705	.	0.00187	-0.00133	-0.00091	-0.00053	175	-2.49861	1	1	0.11764
4	COV	0 Converged	AR_V	AR_V1	.000015926	0.000013921	.	-0.00133	0.00653	0.00212	0.00310	175	-2.49861	1	1	0.11764
5	COV	0 Converged	AR_V	AR_V2	.000015926	0.000009303	.	-0.00091	0.00212	0.00523	0.00172	175	-2.49861	1	1	0.11764
6	COV	0 Converged	AR_V	AR_V3	.000015926	0.000005411	.	-0.00053	0.00310	0.00172	0.00590	175	-2.49861	1	1	0.11764

ADF test for differenced disposable income with 3 lags

Obs	_TYPE_	_STATUS_	_DEPVAR_	_NAME_	_MSE_	Intercept	AR_V	DLAG_V	AR_V1	AR_V2	AR_V3	_NOBS_	_TAU_	_TREND_	_DLAG_	_PVALUE_
1	PARM	0 Converged	AR_V		.000065542	0.005221688	-1	-0.61646	-0.13392	0.023253	-0.024694	175	-5.59192	1	1	.000076853
2	COV	0 Converged	AR_V	Intercept	.000065542	0.000001209	.	-0.00010	0.00007	0.000048	0.000028	175	-5.59192	1	1	.000076853
3	COV	0 Converged	AR_V	DLAG_V	.000065542	-.000100702	.	0.01215	-0.00840	-0.005938	-0.003442	175	-5.59192	1	1	.000076853
4	COV	0 Converged	AR_V	AR_V1	.000065542	0.000068838	.	-0.00840	0.01052	0.006632	0.003184	175	-5.59192	1	1	.000076853
5	COV	0 Converged	AR_V	AR_V2	.000065542	0.000048416	.	-0.00594	0.00663	0.008788	0.004203	175	-5.59192	1	1	.000076853
6	COV	0 Converged	AR_V	AR_V3	.000065542	0.000027901	.	-0.00344	0.00318	0.004203	0.005636	175	-5.59192	1	1	.000076853

estimate cointegrating relationship: C = B1+B2Y+e

The AUTOREG Procedure

Dependent Variable	c

estimate cointegrating relationship: C = B1+B2Y+e

The AUTOREG Procedure

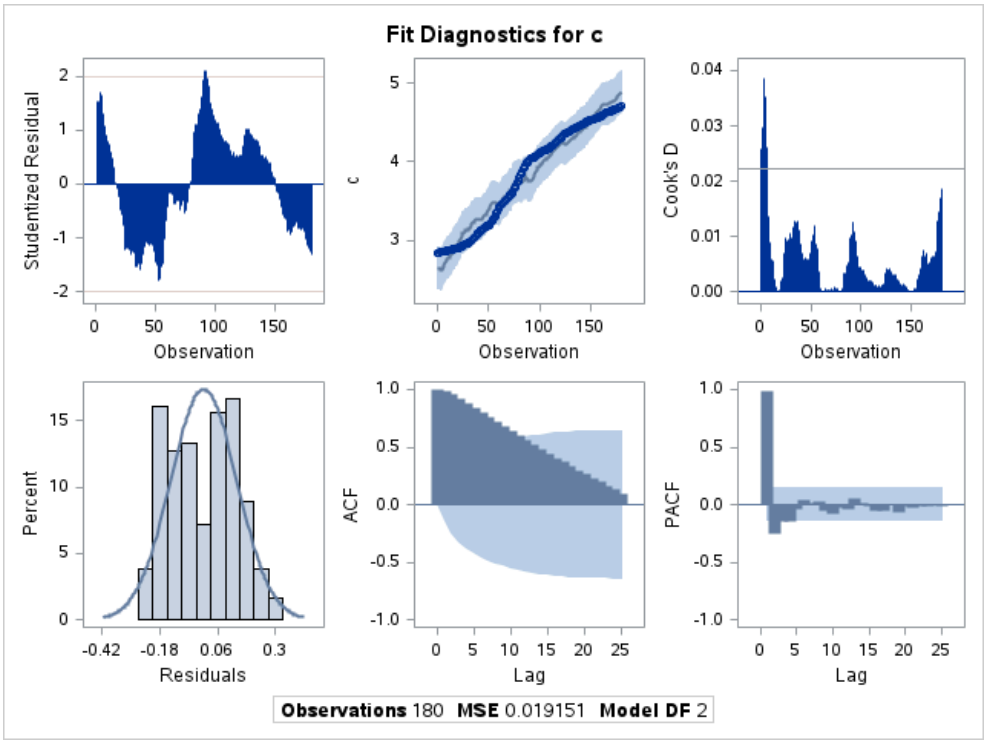
Ordinary Least Squares Estimates			
SSE	3.40885514	DFE	178
MSE	0.01915	Root MSE	0.13839
SBC	-192.78068	AIC	-199.16659
MAE	0.12117118	AICC	-199.09879

Ordinary Least Squares Estimates			
MAPE	3.33997297	HQC	-196.57738
Durbin-Watson	0.0147	Total R-Square	0.9550

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	-2.3273	0.1006	-23.14	<.0001
y	1	1.5318	0.0249	61.48	<.0001

estimate cointegrating relationship: C = B1+B2Y+e

The AUTOREG Procedure



Engle-Granger test for cointegration

The AUTOREG Procedure

Dependent Variable	dehat

Engle-Granger test for cointegration

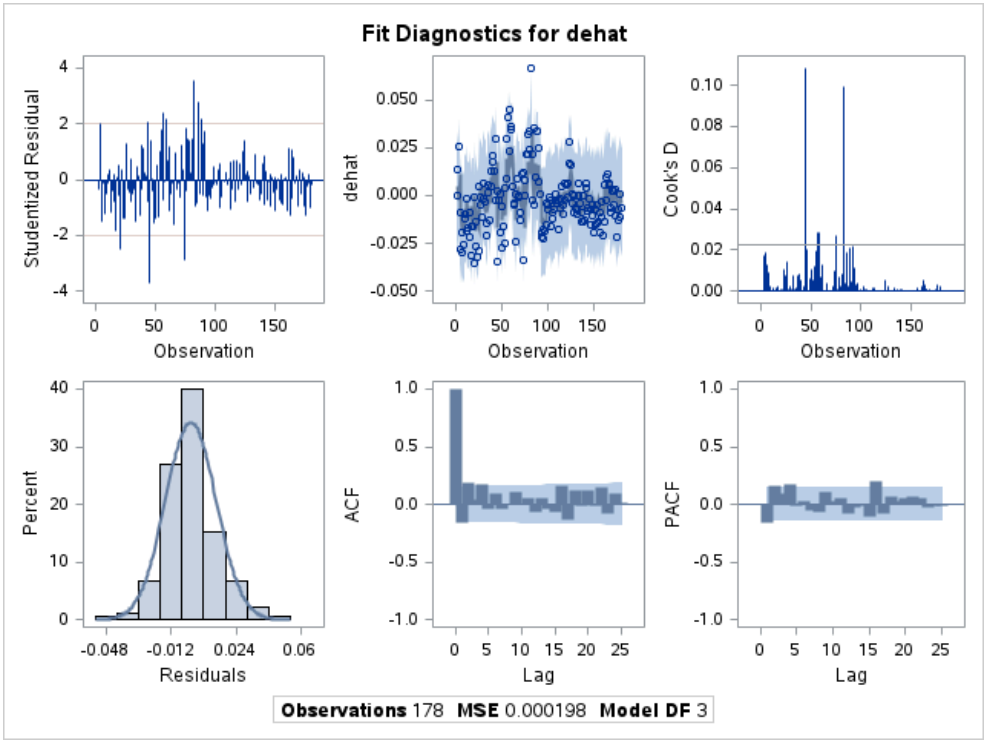
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	0.03488694	DFE	176
MSE	0.0001982	Root MSE	0.01408
SBC	-1004.1562	AIC	-1010.5197
MAE	0.01051014	AICC	-1010.4512
MAPE	522.094835	HQC	-1007.9391
Durbin-Watson	2.3076	Total R-Square	0.2995
NOTE: No intercept term is used. R-squares are redefined.			

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
ehat1	1	-0.0125	0.007715	-1.63	0.1059
dehat1	1	0.5421	0.0631	8.59	<.0001

Engle-Granger test for cointegration

The AUTOREG Procedure



VAR regression for personal consumption expenditure

The AUTOREG Procedure

Dependent Variable	dc

VAR regression for personal consumption expenditure

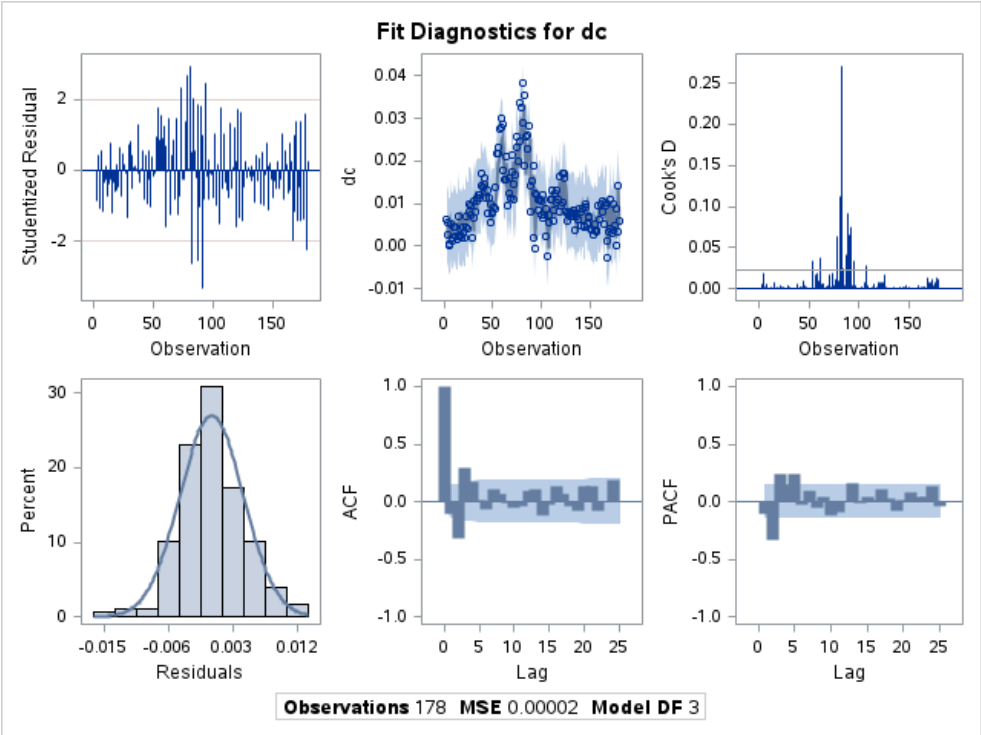
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	0.00347703	**DFE**	175
MSE	0.0000199	**Root MSE**	0.00446
SBC	-1409.4309	**AIC**	-1418.9763
MAE	0.0034163	**AICC**	-1418.8383
MAPE	83.9845911	**HQC**	-1415.1054
Durbin-Watson	2.1946	**Total R-Square**	0.6673

Parameter Estimates					
Variable	**DF**	**Estimate**	**Standard Error**	**t Value**	**Approx Pr > |t|**
Intercept	1	0.001433	0.000710	2.02	0.0453
dc1	1	0.8268	0.0447	18.49	<.0001
dy1	1	0.0464	0.0399	1.17	0.2455

VAR regression for personal consumption expenditure

The AUTOREG Procedure



VAR regression for real personal disposable income

The AUTOREG Procedure

Dependent Variable	dy

VAR regression for real personal disposable income

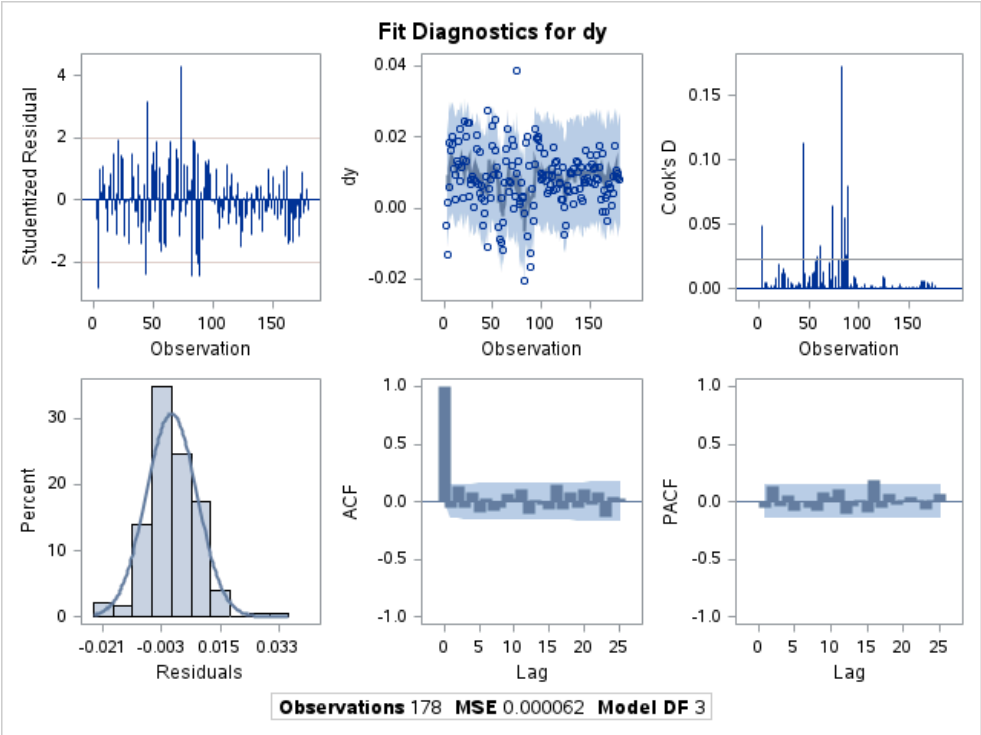
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	0.01077998	DFE	175
MSE	0.0000616	Root MSE	0.00785
SBC	-1208.0215	AIC	-1217.5669
MAE	0.00586469	AICC	-1217.4289
MAPE	309.712684	HQC	-1213.696
Durbin-Watson	2.1103	Total R-Square	0.1688

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	0.009814	0.001251	7.85	<.0001
dc1	1	-0.3269	0.0787	-4.15	<.0001
dy1	1	0.2285	0.0702	3.26	0.0014

VAR regression for real personal disposable income

The AUTOREG Procedure



estimate VAR model and generate impulse response

The VARMAX Procedure

Number of Observations	179
Number of Pairwise Missing	1

Simple Summary Statistics

Variable	Type	N	Mean	Standard Deviation	Min	Max
dc	Dependent	179	0.01045	0.00767	-0.00282	0.03822
dy	Dependent	179	0.00819	0.00859	-0.02039	0.03866

estimate VAR model and generate impulse response

The VARMAX Procedure

Type of Model	VAR(1)
Estimation Method	Least Squares Estimation

Model Parameter Estimates

Equation	Parameter	Estimate	Standard Error	t Value	Pr > |t|	Variable
dc	CONST1	0.00143	0.00071	2.02	0.0453	1
	AR1_1_1	0.82682	0.04471	18.49	0.0001	dc(t-1)
	AR1_1_2	0.04645	0.03986	1.17	0.2455	dy(t-1)
dy	CONST2	0.00981	0.00125	7.85	0.0001	1
	AR1_2_1	-0.32695	0.07872	-4.15	0.0001	dc(t-1)
	AR1_2_2	0.22851	0.07018	3.26	0.0014	dy(t-1)

Covariances of Innovations

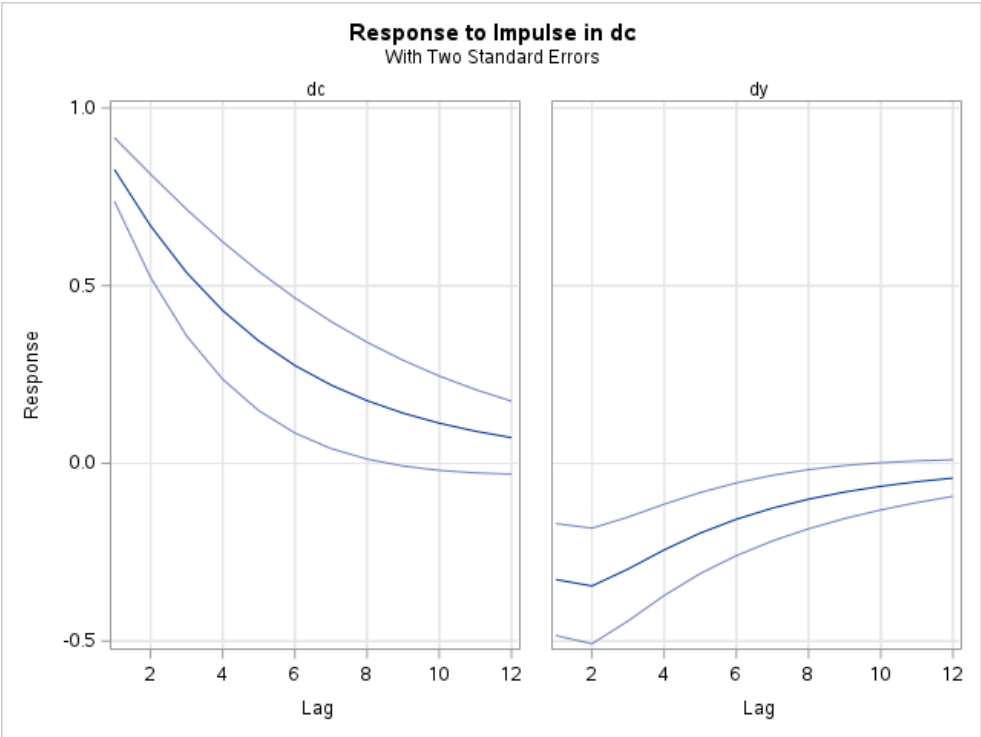
Variable	dc	dy
dc	0.00002	0.00000
dy	0.00000	0.00006

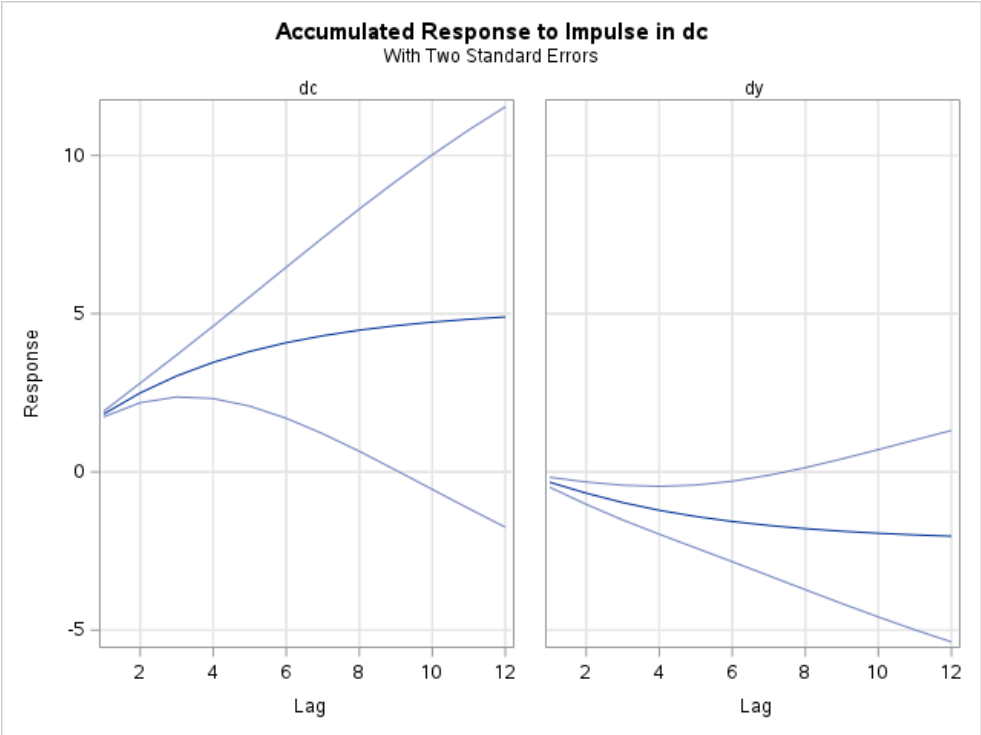
Log-likelihood	1652

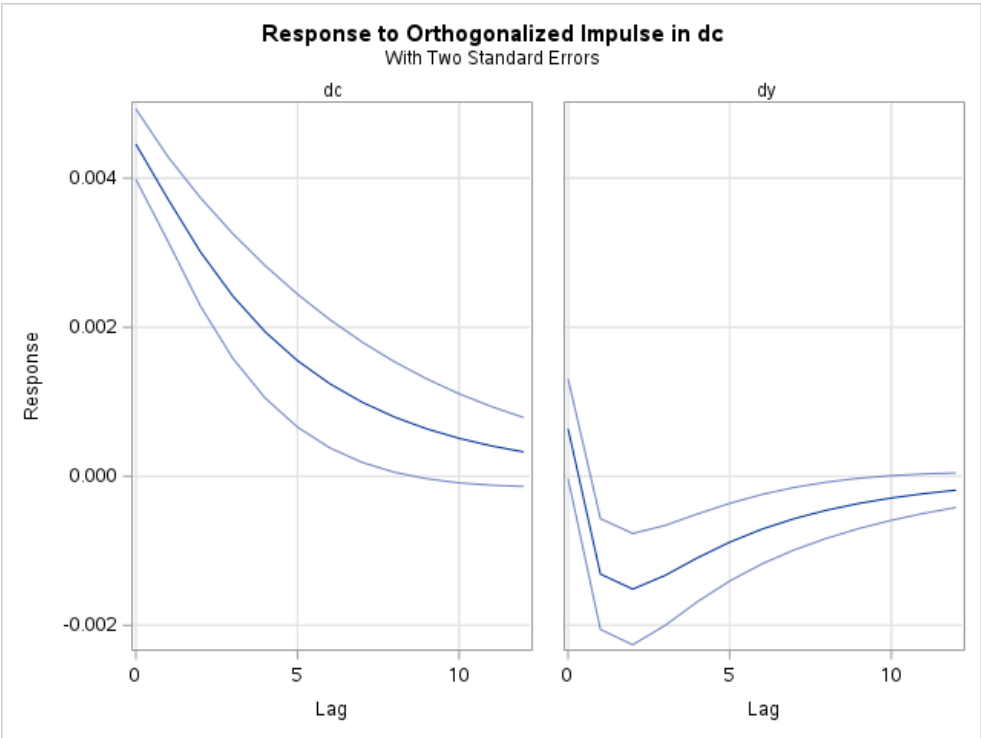
Information Criteria	
AICC	-3284.93
HQC	-3274.39

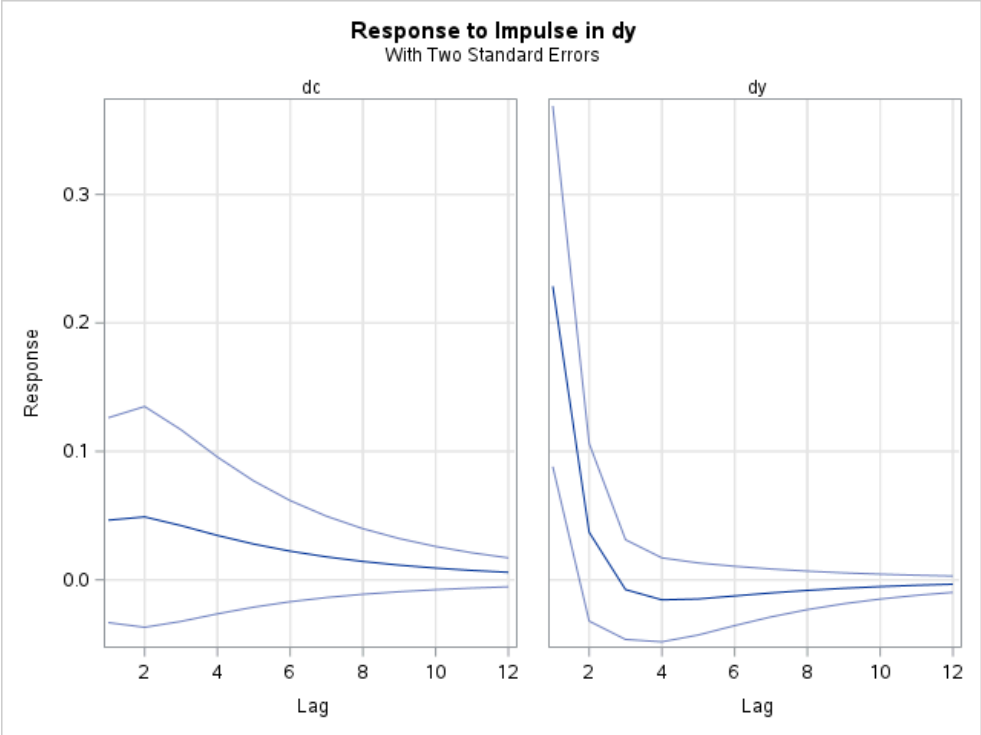
Information Criteria	
AIC	-3286
SBC	-3257.36
FPEC	1.257E-9

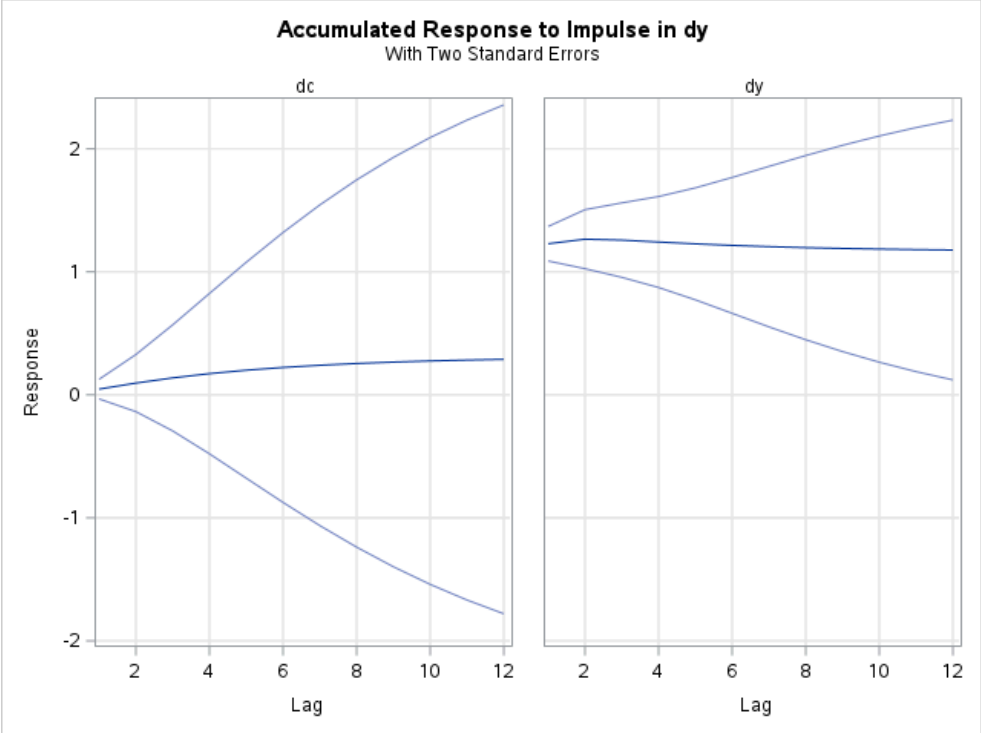
Forecasts					
Variable	Obs	Forecast	Standard Error	95% Confidence Limits	
dc	180	0.00674	0.00446	-0.00200	0.01547
	181	0.00745	0.00581	-0.00394	0.01885
	182	0.00805	0.00656	-0.00480	0.02091
	183	0.00854	0.00700	-0.00518	0.02225
	184	0.00893	0.00727	-0.00531	0.02317
	185	0.00924	0.00743	-0.00533	0.02381
dy	180	0.00972	0.00785	-0.00566	0.02511
	181	0.00983	0.00816	-0.00615	0.02582
	182	0.00962	0.00830	-0.00664	0.02589
	183	0.00938	0.00841	-0.00710	0.02586
	184	0.00917	0.00848	-0.00745	0.02578
	185	0.00899	0.00853	-0.00772	0.02570

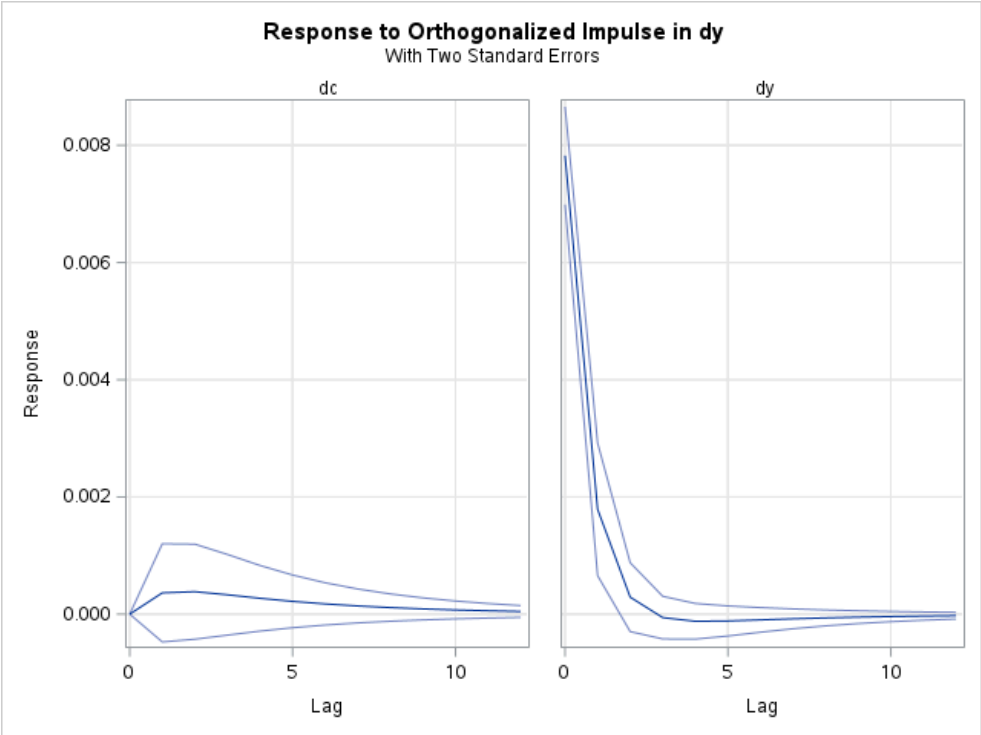



Accumulated Response to Impulse in dc
With Two Standard Errors
dc
dy
Response
10
5
0
-5
2
4
6
8
10
12
Lag


Response to Orthogonalized Impulse in dc
With Two Standard Errors
dc
dy
Response
0.004
0.002
0.000
-0.002
0
5
10
Lag


Response to Impulse in dy
With Two Standard Errors
dc
dy
Response
0.3
0.2
0.1
0.0
2
4
6
8
10
12
Lag


Accumulated Response to Impulse in dy
With Two Standard Errors
dc
dy
Response
2
1
0
-1
-2
2
4
6
8
10
12
Lag


Response to Orthogonalized Impulse in dy
With Two Standard Errors
dc
dy
Response
0.008
0.006
0.004
0.002
0.000
0
5
10
Lag
0
5
10
Lag